

KALIGITHI PJ PAUL

DATA ENGINEERING | DATABRICKS | CLOUD

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PROFESSIONAL SUMMARY

Generative AI (GenAI) and data engineering professional with hands-on internship experience building enterprise Databricks applications that combine LLM reasoning via Databricks Model Serving (Mosaic AI), governed metadata, SQL analytics, and API services. Skilled in Python, SQL, FastAPI, Streamlit, Databricks SQL Warehouses, System Tables, AWS, containers, and GitLab CI. Built job-failure analysis and FinOps assistants with structured recommendations, cache-first LLM workflows, and safety controls for troubleshooting and cost optimization. Actively seeking full-time Data Engineering / GenAI Engineering opportunities.

TECHNICAL SKILLS

Programming: Python, SQL, Java, Machine Learning

Generative AI (GenAI): LLM applications, Databricks Model Serving (Mosaic AI), prompt engineering, RAG, semantic search, vector search, agentic AI, structured JSON, caching, AI recommendations

Databricks: SQL Warehouses, Jobs, Workflows, Clusters, Unity Catalog, Delta Lake, Model Serving (Mosaic AI), System Tables, billing usage, DBU analysis, FinOps, DBT, Power BI, Tableau

Data and Backend: Apache Spark fundamentals, ETL/ELT, Data pipelines, FastAPI, REST APIs, Streamlit, microservices, API integration

Cloud: AWS EC2, IAM, S3, Linux

DevOps: Docker, Git, GitHub, containerization, monitoring, observability

PROFESSIONAL EXPERIENCE

Amgen - GenAI Analyst Intern

Jan 2026 - Jul 2026

Enterprise GenAI and Databricks Engineering

- Built an AI-powered Databricks job-failure analysis platform that combines workflow context and enterprise metadata with LLM reasoning (Databricks Model Serving / Mosaic AI) to produce structured root-cause explanations and remediation recommendations.
- Developed a **Streamlit** interface and FastAPI REST services backed by **Databricks SQL Warehouse**, automating failed-job analysis and returning structured JSON for downstream workflows.
- Designed cache-first request handling to reduce redundant LLM calls and improve repeat-analysis efficiency; grounded recommendations in job and platform metadata.
- Containerized application services with Docker and integrated AWS EC2, IAM, and S3 into the delivery environment.

PROJECTS

Databricks FinOps AI Assistant *Python | SQL | Databricks System Tables | Unity Catalog | FastAPI | LLMs*

Mar 2026 - Jul 2026

- Built a natural-language FinOps assistant that analyzes SQL Warehouse utilization and **DBU** consumption, then generates evidence-backed **cost-optimization** recommendations.
- Generated schema-grounded read-only **SQL**, validated access against approved Unity Catalog tables, blocked unsafe requests, and used intent-based caching with optimized System Table queries.

Integrated Player Performance Analysis – Football Dataset (Dec 2025)

- Built a **Python-based** system using **machine learning** to analyze football player performance by position, injury risk, and fatigue.
- Predicted injury risk with **XGBoost** and analyzed fatigue using **regression** to support smarter player rotation and workload management.
- Visualized key insights with **Power BI** and **Seaborn** to help coaches make data-driven team and strategy decisions.

Enhancing E-Commerce Business Insights through Predictive and Prescriptive Analytics (Aug 2024)

- Processed and analyzed large-scale e-commerce data (8,500+ transactions, 12+ features) using Python (Pandas, NumPy, Matplotlib, Seaborn); identified data inconsistencies and prepared clean dataset for analytics.
- Designed business intelligence dashboards across **6+ KPI** categories highlighting sales performance by outlet type and product category; extracted insights driving 60%+ revenue concentration.

EDUCATION

- B.Tech in Computer Science and Engineering** — Vellore Institute of Technology, Andhra Pradesh | CGPA: 7.72/10 | 2022-2026
- Class XII: 86.6% — Sasi Educational Institutes, Velivenu | 2022
- Class X: 84.8% — Sasi English Medium High School | 2020

CERTIFICATIONS AND ACHIEVEMENT

Certifications:

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional, Oracle University | 2025
- Databricks Fundamentals Accreditation, Databricks Academy | 2026
- DBT Learn: Fundamentals, DBT Labs
- Databricks Academy: AI/BI for Data Analysts | 2026

Achievement: Ranked in the top 4% among 100,000 participants in a state-level engineering entrance examination.